CBSE Class 10 Science Previous Year Question Paper – Set 31/1/1

Time Allowed: 3 hours
Maximum Marks: 80

General Instructions

All questions are compulsory.
Internal choice is provided wherever applicable.
Questions follow the current CBSE syllabus.
Do not create image-based or map-based questions.

SECTION A – Objective Questions

1. **Which of the following solutions will turn blue litmus red?**
(a) Baking soda solution
(b) Washing soda solution
(c) Vinegar
(d) Sodium chloride solution

2. **Which metal is used to galvanize iron to prevent rusting?**
(a) Copper
(b) Zinc
(c) Aluminium
(d) Magnesium

3. **Which of the following is a non-renewable source of energy?**
(a) Wind
(b) Solar
(c) Coal
(d) Tidal

4. **Which change represents a chemical reaction?**
(a) Dissolving salt in water
(b) Melting of ice
(c) Burning of paper
(d) Breaking glass

5. **What is the ratio of hydrogen to oxygen by mass in water?**
(a) 1:2
(b) 2:1
(c) 8:1
(d) 1:8

6. **The process of converting nitrogen gas into ammonia is called**

(a) Nitrogen fixation
(b) Nitrification
(c) Ammonification
(d) Denitrification

7. **Which part of the human eye controls the amount of light entering it?**
(a) Lens
(b) Iris
(c) Cornea
(d) Retina

8. **Budding is a mode of reproduction seen in**
(a) Yeast
(b) Amoeba
(c) Plasmodium
(d) Hydra

9. **Which of the following is used as a preservative in pickles?**
(a) Vinegar
(b) Baking powder
(c) Caustic soda
(d) Dry ice

10. **Which of these is a characteristic of metals?**
(a) Poor conductor of heat
(b) Dull appearance
(c) Malleability
(d) Brittle nature

11. **What is the chemical formula of calcium hydroxide?**
(a) CaO
(b) $Ca(OH)_2$
(c) $CaCO_3$
(d) $CaCl_2$

12. **DNA copying during cell division is important because it**
(a) Prevents variation
(b) Ensures continuity of traits
(c) Reduces cell number
(d) Increases cell size

13. **The main organs for regulating water in plants are**
(a) Roots
(b) Flowers
(c) Stomata
(d) Seeds

14. **Saponification is related to the preparation of**
(a) Salt
(b) Soap
(c) Glass
(d) Alcohol

15. **The refractive index of a medium is always**
(a) Less than one
(b) More than one
(c) Equal to one
(d) Zero

16. **Which of these structures in a cell contains genetic material?**
(a) Cell wall
(b) Nucleus
(c) Mitochondria
(d) Cytoplasm

17. **Which of these does not describe the periodic classification of elements?**
(a) Elements are arranged according to atomic mass
(b) Elements show periodicity of properties
(c) Groups have similar properties
(d) Elements in a period increase in atomic number

18. **The product of fertilization in plants develops into**
(a) Seed
(b) Leaf
(c) Fruit
(d) Root

19. **When carbon dioxide is passed through lime water, it turns**
(a) Blue
(b) Yellow
(c) Milky
(d) Colourless

20. **Direction – Read the following Assertion and Reason and answer as instructed.**
Assertion (A): Green plants are called autotrophs.
Reason (R): Green plants make their own food from simple substances.
Choose the correct option:
(a) Both A and R are true, and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, R is false
(d) A is false, R is true

SECTION B – Short Answer Questions (2 marks)

21. **Write two differences between biodegradable and non-biodegradable substances.**

22. **Why is the use of excessive fertilizers and pesticides harmful to the environment?**

23. **List two characteristics of homologous series with one example.**

24. **Why are decomposers important for the ecosystem?**

25. **Explain the role of ozone in the atmosphere.**

26. **Why does a ray of light bend when it passes from air into water?**

27. **List two advantages of using solar panels for home energy.**

28. **State two observations when iron nails are dipped in copper sulphate solution.**

29. **Give two differences between respiration and combustion.**

30. **State two disadvantages of thermal power plants.**

31. **How do Mendel's experiments show that traits are inherited independently?**

32. **Explain why saline hydra does not reproduce by binary fission.**

33. **Give two reasons for the increase in environmental pollution.**

34. **What is meant by 'reflex action'? Give one example.**

35. **Why is it not possible for us to see clearly through fog?**

36. **List two differences between metals and non-metals.**

37. **How is carbon dioxide utilized in the process of photosynthesis?**

38. **State any two precautions while using electrical appliances at home.**

39. **Why is it recommended to use LEDs instead of incandescent bulbs?**

40. **Name two environmental consequences of deforestation.**

SECTION C – 3-Mark Questions

41. **Explain with examples how chemical reactions can be endothermic and exothermic in nature.**

42. **A student adds dilute hydrochloric acid to sodium carbonate. Write the balanced chemical equation and mention the observations.**

43. **Describe the steps involved in the carbon cycle in nature.**

44. **What are the main differences between voluntary and involuntary actions in animals?**

45. **Explain the function of the human kidney and state its importance.**

46. **Why are the leaves of some plants modified into tendrils? Mention their advantage.**

47. **Why does the image formed by a plane mirror appear laterally inverted?**

48. **Explain the need and significance of sustainable management of natural resources.**

SECTION D – Case Study / Source-Based Questions

49. **Read the following case and answer the questions that follow:**
A group of students visited a dam and observed water being released through turbines to generate electricity. They also noticed the surrounding area being lush green due to availability of water. However, the local people shared concerns about displacement and loss of habitats.

(a) Why is hydroelectricity considered renewable?
(b) Name one disadvantage of constructing large dams.
(c) How does release of water from dams help in irrigation?

50. **Read the passage and answer the following questions:**
During a biology class, a teacher explains that insulin regulates blood sugar. A student asks why the pancreas is called both an exocrine and endocrine gland.

(a) What is the endocrine function of pancreas?
(b) What is its exocrine function?
(c) How does insulin deficiency affect the body?

SECTION E – Long Answer Questions (5 marks)

51. **Describe in detail the process of digestion in the human body, mentioning the role of each major organ involved. Highlight the enzymatic actions that take place in different parts of the digestive tract and explain how nutrients are absorbed and assimilated.**

52. **Electric circuits are a crucial part of daily life. Discuss the differences between series and parallel connections in electrical circuits, their advantages and disadvantages, and give practical examples from household use to support your analysis.**

Answer Key – CBSE Class 10 Science – Set 31/1/1

1. (c)
2. (b)
3. (c)
4. (c)
5. (d)
6. (a)
7. (b)
8. (a)
9. (a)
10. (c)
11. (b)
12. (b)
13. (c)
14. (b)
15. (b)
16. (b)
17. (a)
18. (a)
19. (c)
20. (a)

21. Biodegradable substances can be broken down by natural processes while non-biodegradable cannot; for example, paper is biodegradable, plastic is not.
22. Excessive fertilizers and pesticides contaminate soil and water, harming organisms and disrupting ecosystems.
23. Homologous series have the same functional group and a difference of CH_2 between members; e.g., alkanes.
24. Decomposers recycle nutrients back into the ecosystem and help in breaking down dead matter.
25. Ozone absorbs harmful ultraviolet radiation, protecting living organisms from DNA damage.
26. Due to change in speed and direction as light moves from one medium to another because of different densities.
27. Solar panels reduce dependency on fossil fuels and lower household electricity bills.
28. The blue color of copper sulphate fades, and a reddish-brown layer forms on the iron nails.
29. Respiration occurs in living organisms and is a slow process; combustion is fast and occurs outside cells.
30. They release greenhouse gases and cause depletion of natural resources.
31. Mendel's dihybrid cross showed independent assortment of traits; seed color and shape inherited separately.

32. Hydra reproduces by budding, not fission, as it forms new outgrowths while binary fission splits organisms equally.
33. Increased industrialization and rapid population growth.
34. Reflex action is an involuntary, automatic response to a stimulus; e.g., pulling hand away from a hot object.
35. Fog consists of tiny water droplets that scatter light, reducing visibility and clarity.
36. Metals are malleable and good conductors, whereas non-metals are brittle and poor conductors.
37. Plants use carbon dioxide during photosynthesis to make glucose, releasing oxygen.
38. Do not touch with wet hands; ensure proper earthing of appliances.
39. LEDs consume less power and last longer compared to incandescent bulbs.
40. Loss of biodiversity and increased soil erosion.
41. Endothermic reactions absorb heat, like photosynthesis, while exothermic reactions release heat, such as burning of fuel. For example, dissolving ammonium chloride in water feels cold (endothermic) and combustion of wood releases heat (exothermic). These differences depend on energy changes during the reaction.
42. $Na_2CO_3 + 2HCl \rightarrow 2NaCl + H_2O + CO_2$. Observation: Effervescence due to carbon dioxide gas and the solution may turn cloudy.
43. The carbon cycle involves respiration releasing CO_2, photosynthesis absorbing it, decomposition of dead organisms returning carbon to the soil, and combustion adding CO_2 back to the atmosphere. This maintains Earth's carbon balance.
44. Voluntary actions are under conscious control, like moving a hand, while involuntary actions occur automatically, such as heartbeat. Voluntary actions involve brain, involuntary involve spinal cord or brainstem.
45. Kidneys filter blood to remove waste, produce urine, and regulate water and salt balance, maintaining homeostasis and overall health.
46. Tendrils help climbing plants attach to supports and access sunlight. They provide structural advantage in dense vegetation.
47. Lateral inversion occurs because mirrors reflect light in such a way that left and right are reversed while top and bottom remain the same.
48. Sustainable management ensures resources are available for the future and maintains ecological balance, preventing depletion and environmental damage.
49. (a) Hydroelectricity is renewable as it uses water, a natural cycle resource. (b) Large dams can cause displacement and loss of habitats. (c) Water released helps irrigation by supplying regular water to fields.
50. (a) Endocrine function is secretion of hormones like insulin. (b) Exocrine function is secreting digestive enzymes. (c) Insulin deficiency leads to high blood sugar and diabetes.
51. Food is chewed and mixed with saliva in the mouth, digested with enzymes. Stomach uses acid and enzymes to break proteins, then small intestine further digests with pancreatic juices and bile. Absorption happens through intestinal walls, nutrients are carried via blood for assimilation, while undigested food goes to the large intestine.
52. Series circuits have one path for current, parallel circuits have multiple paths. In series, if one

component fails, all stop working; parallel allows independent functioning. Parallel circuits are common in homes for controlled lighting and appliances. Series circuits are used in simple devices like torches. Each has unique advantages for specific applications.